EXP: 04 Setting Up a Local Area Network (LAN)

DATE : 10/8/26

1. Aim:

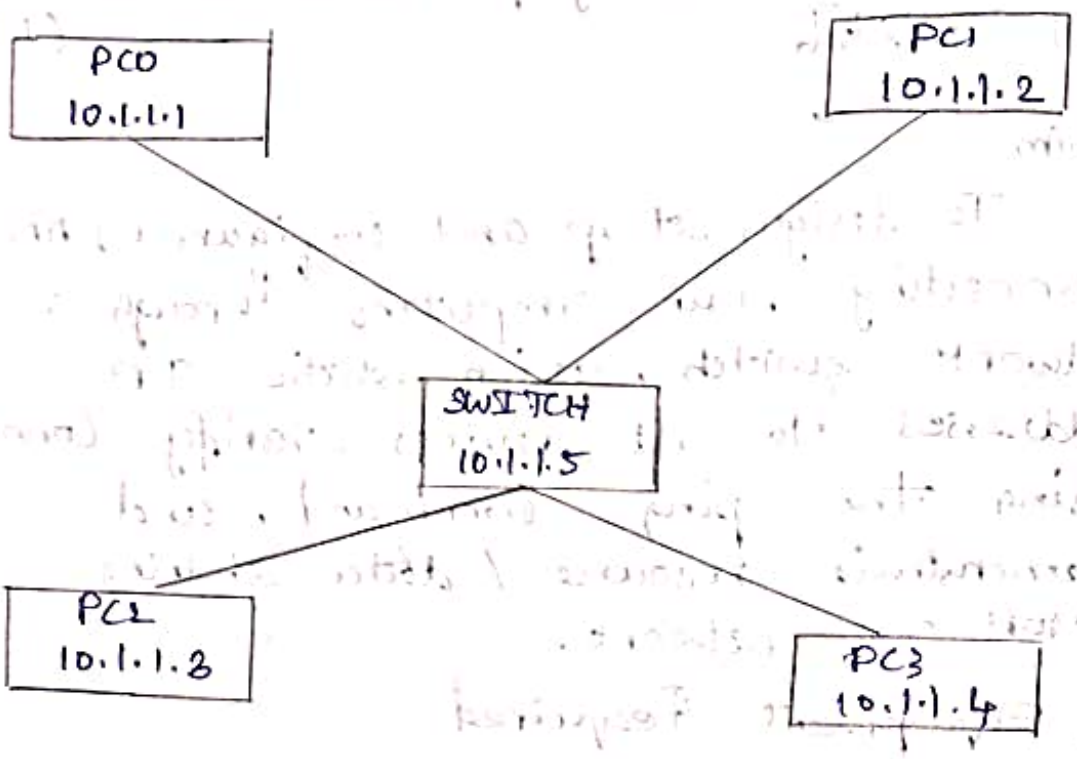
To design, set up and configure a LAN connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network.

2. Equipment Required

S.No	Equipment	Quantity	Specification
1.	Personal Computers (PC)	4	Ethernet (NIC) enabled
2.	Network switch	1	8/16/24 port, unmanaged / managed
3.	Ethernet Cables (Cat 5e/Cat 6)	4	Straight-through, RJ-45 Connectors.

4. Network Diagram

The LAN was implemented using a Star Topology as shown below, with 4 PCs connected individually to a central switch.



5. IP Address Configuration Table

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa
PC1	10.1.1.2	255.0.0.0	Switch Port Fa
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0
PC3	10.1.1.4	255.0.0.0	Switch Port Fa
Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1, Intaf

6. Implementation in CISCO Packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using CISCO packet tracer simulation software, following the steps below.

Step 1: Place the Devices

1. Open Cisco packet tracer.
2. From the device Category panel, select End Devices and drag 4 PCs onto the workspace.
3. Select switches and drag one switch onto the workspace.

Step 2: Cable the devices

4. Click connections → select Copper Straight-Through Cable.
5. Click each PC, then click the Switch and select an available FastEthernet port to complete each connection.
6. Repeat for all four PCs. Green link lights confirm successful.

Step 3: Assign IP Address to PCs

7. Click a PC → Desktop tab → IP Configuration.
8. Enter the static IP Address and Subnet Mask as per the IP table for each PC.

Step 4: Configure the switch management IP (optional)

9. Click the switch → CLI tab

10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface Vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify Connectivity using Ping

11. Click a PC → Desktop tab → Command prompt.

12. Use the ping command to test connectivity to the other PCs.

13. Successful "reply from ..." message confirm connectivity.

Step 6: Demonstrate Data sharing (Server Simulation)

14. Add a server device (End devices → Server-PT) and connect it to the switch.

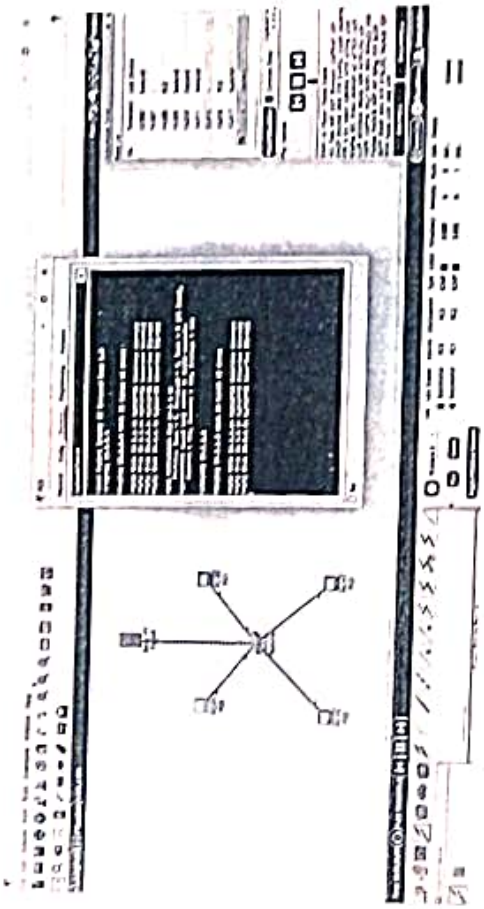
15. Assign it an IP address (eg. 10.1.1.10)

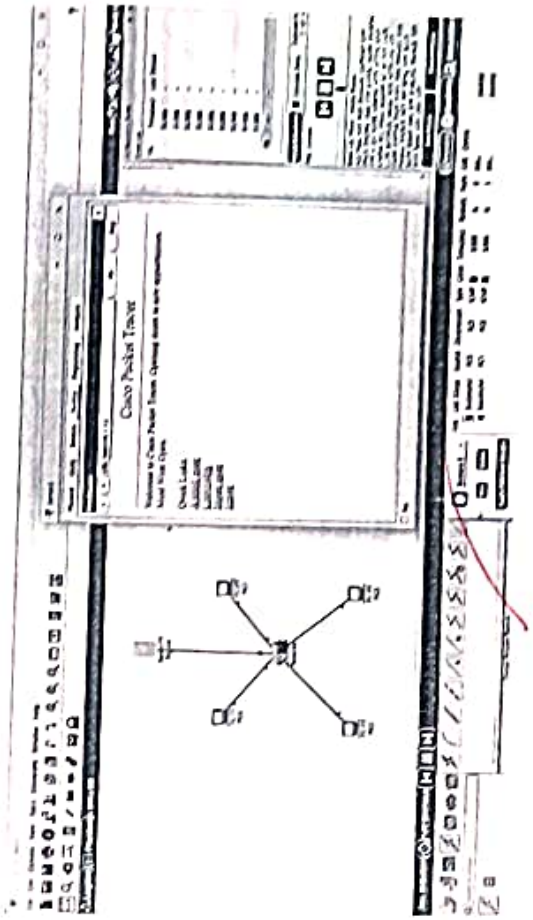
16. Open the server's services (Config) tab and enable the HTTP Service.

Step 7: Observe Packet flow in simulation mode

17. Switch to simulation mode (bottom-right of the workspace).

18. Send a ping or web request between two devices and use the Play/Capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.





RESULT:

Thus the LAN is designed and set up, and connection is verified.

mode enabled - its enabled
default you also see all t
packets on the network in
only packets addressed to
network.